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Training and transfer effects in task-switching

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Abstract

Performance on task-switching, a paradigm commonly used to measure executive function, has been shown to improve with practice. However, no study to date has tested whether these benefits are specific to the tasks learned or are transferable to new situations. We report evidence of transferable improvement in a cued randomly switching paradigm as measured by mixing cost, but no consistent improvement for switch cost. Improvement in mixing costs arises from a relative reduction in time to perform both switch and non switch trials that immediately follow switch trials, implicating the ability to recover from unexpected switches as the source of improvement. These results add to a growing number of studies demonstrating generalizable improvement with training on executive processing.

Training and transfer effects in task-switching

Task-switching is an increasingly popular method to study cognitive control. In the typical task-switching paradigm, a subject must switch between two simple tasks such as adding three to a number on one trial and then subtracting three on the next. When subjects alternate between two tasks their reaction times (RTs) are slower than when performing one task alone.

Although the task-switching paradigm was originally developed by Jersild (1927), its use has only recently become widespread due to a growing interest in executive functions. Executive functions are supervisory processes that allow for top down control of action and are essential for goal-directed behavior. Task-switching has been proposed as a candidate executive function along with the maintenance and updating of information in working memory, inhibition, and the ability to perform two tasks at the same time (Miyake et al., 2000). Moreover, it is used to study executive processing across psychology and related fields including cognitive development (Cepeda, Kramer, & Gonzalez de Sather, 2001), cognitive aging (Kramer, Hahn, & Gopher, 1999; Kray & Lindenberger, 2000, Mayr, 2001), brain imaging studies (Braver, Reynolds, & Donaldson, 2003; Sohn, Ursu, Anderson, Stenger, & Carter, 2000) and in a wide array of clinical disorders including attention deficit hyperactivity disorder (Cepeda, Cepeda, & Kramer, 2000), Parkinson's disease (Meiran, Friedman, & Yehene, 2004) and frontal lobe injury (Keele & Rafal, 2000).

Given the importance of executive functions and their impairment in numerous populations, there is growing interest in determining whether performance on executive control can be improved with practice (Verhaeghen, Cerella, & Basak, 2004) and show transfer to unpracticed situations. There is some evidence that executive functions such as inhibition (Dowsett & Livesey, 2000), working memory (Klingberg, Forssberg, & Westerberg, 2002) and dual task performance (Bherer et al., 2005; Kramer, Larish, & Strayer, 1995) not only become more efficient with practice, but that this improvement can transfer to novel contexts. There are demonstrable practice-related improvements in switching performance (Jersild, 1927; Kray & Eppinger, 2006; Kray & Lindenberger, 2000; Kramer et al., 1999; Mayr & Kliegl, 2003; Meiran, 1996). However, there has been no direct test of whether the better performance seen constitutes a transferable improvement in the ability to switch tasks more efficiently or simply an improvement in the specific tasks practiced. In the current study, we investigated the extent to which improved task-switching performance with practice can transfer to a new task-switching situation. We employed a pre-test/post-test design in which participants were measured on their ability to switch between two tasks both before and after practice on other switching tasks.

Task-switching and improvements with practice

Cognitive control has been proposed as necessary on a number of different levels when switching between two tasks. Participants must select and maintain the currently relevant goal, inhibit currently irrelevant aspects of stimulus and suppress stimulus-response mappings belonging to the competing task. Rogers and Monsell (1995) introduced the term task-set reconfiguration to describe the proposed set of control processes necessary when switching from one task to another. Task-set reconfiguration is measured by the switch cost, the average RT of the switch trials is compared to the average of the non-switch trials within a switch block. However, other researchers view the difference between switch and non-switch trials as arising less from the cost of a switch and more from a repetition benefit for non-switch trials due to various forms of priming (Wylie & Allport, 2000; Dreisbach, Haider, & Kluwe, 2002; Sohn & Carlson, 2000) or cue encoding and retrieval benefits (Mayr & Kliegl, 2003; Schneider & Logan, 2005). Of these two interpretations, only task-set reconfiguration allows for the possibility of transferable improvement in task-switching as measured by the switch cost while priming effects and cue-based retrieval processes would seem unlikely to generalize from one set of tasks to another.

A different perspective is that cognitive control is necessary on both the switch and the non-switch trials within a switching block. Although there is no explicit task-set reconfiguration needed on non-switch trials, the stimulus may still evoke the currently irrelevant task set which must be inhibited for the

moment, but not completely abandoned as it will be needed on future trials.

Therefore, some task-switching research also uses mixing cost as a measure of cognitive control. Mixing cost is the comparison of the average RT of trials from single-task blocks to the average RT of non-switch trials within a switching block. Proposed sources of mixing cost include the control processes needed to resolve stimulus ambiguity (Rubin & Meiran, 2005) and to maintain and select between multiple task sets (Mayr, 2003). Such processes may include the ability to focus attention on the relevant aspects of the stimulus and the response set and to inhibit irrelevant ones. However, others view mixing cost as simply reflecting differences in arousal and working memory load between single-task and switching blocks (Rogers & Monsell, 1995).

Both mixing and switch cost show substantial reduction with practice (Cepeda et al., 2001; Kray & Lindenberger, 2000; Meiran, 1996). On the level of the individual trial types, this means that non-switch trials show greater practice-based reductions than single-task trials and switch trials show larger reductions than non-switch trials. However, whether the same pattern will be present in any possible transfer remains to be seen.

Experiment 1

This experiment was designed to establish the extent to which practice-based reductions in two measures of switching cost can transfer to a new switching situation. We used a pre-test/post-test design in which switching

between a pair of transfer tasks was measured before and after training on either switching using three task pairs different from the pre-test/post tasks or single-task practice on the same training tasks. In addition, we used two different variants of the task-switching paradigm, predictable and random switching.

In predictable switching the participant has full knowledge of the task-switching sequence (e.g. the two tasks alternate every two trials) so that he or she knows the identity of the next trial type, switch or non-switch, before it happens. In random switching, the task to be performed changes in an unpredictable fashion, and the appropriate task on each trial must be indicated by a cue. Both predictable and random switching require participants to inhibit irrelevant aspects of the stimulus and the inappropriate response set on any given trial as well as task-set reconfiguration on switch trials. However, differences between predictable and random switches have been reported for both mixing (Kray, Li & Lindenberger, 2002) and switch costs (Milán, Sanabria, Tornay, & González, 2005; Monsell, Sumner, & Waters, 2003; Tornay & Milán, 2001), leading some researchers to propose that the demand for controlled processing is greater in random switching than in predictable switching (Tornay & Milán, 2001) while others have concluded the opposite (Milán et al., 2005; Monsell et al., 2003). Therefore, we included a block of predictable switching and a block of random switching at pre-test and post-test. We had three separate groups of participants, one that trained on switching predictably, a second that trained on random

switching and a third group that practiced the same tasks, but in single-task blocks that did not require them to switch between tasks. This allowed us to test not only whether transfer of training was possible for two types of task-switch, but also whether practice on one would transfer to the other.

Method

Participants. Ninety-three University of Michigan undergraduates participated for a \$20 payment and were randomly assigned to the predictable switch-training group ($N = 31$), the random switch-training group ($N = 31$), or the control group ($N = 31$).

Design. To test for transferable improvement in switching performance, we employed a pre-test/post-test design in which all three groups were initially measured on both a predictable block of switching and a random block of switching. The experiment was conducted across two days. On day one, pre-test switching performance was measured after which, each group was given a different course of training. One group practiced switching between three different pairs of tasks using predictable switches. Another practiced switching between the same pairs of tasks, but with random switches, and a third group practiced performing each of the six tasks in isolation. On day two, all three groups continued their particular course of training followed by the post-test measurement on predictable and random switching. Transfer of training would be indicated by differential pre-test to post-test improvement between groups.

Tasks and Stimuli. Four pairs of tasks were used in this experiment, one pair was used at pre and post-test and three were used during training. All tasks required simple judgments about numbers or letters and used the same two response keys designated the “left” and the “right” keys. In predictable switching, the task to be performed switched every other trial (AABBAA) while in random switching the two tasks were allowed to change at random with the constraint that half the trials were switches so that both the predictable and random switching conditions received the same number of switch and non-switch trials. Highly transparent verbal cues that indicated the nature of the task and the appropriate response mapping were present in the upper left and right hand corners of the screen on each trial. These cues were used to reduce working memory demands and were present for all three possible block types: single-task blocks, predictable switching blocks and random switching blocks. The cues appeared simultaneously with each stimulus and there was no delay between a response and the subsequent trial. Bivalent stimuli (stimuli relevant to either task) were present even in the single task blocks. All tasks were programmed in E-Prime 1.1 (Schneider, Eschman & Zuccolotto, 2002).

In the Consonant-Vowel/Odd-Even transfer task pair (CVOE), participants saw a letter-number pair in the center of the screen and determined whether the letter was a consonant or a vowel or whether the number was odd or even. Participants first completed two single-task blocks, one of solely Consonant-

Vowel trials and another of Odd-Even trials. They were then given two blocks of task-switching: a block consisting of predictable switching and a block of random switching. The order of these blocks was counterbalanced across participants. The cues, Consonant or Odd and Vowel or Even were present in the left and right-hand corners of the screen. These cues were present for all four blocks of trials, the two single-task blocks as well as the two switching blocks.

Training consisted of repeated practice on three different pairs of tasks. Three different task-switching training pairs were used in light of work in the skill acquisition literature indicating that variability in training promotes greater transfer than training using only one task (see Schmidt & Bjork, 1992 for a review).

The task-switching training tasks were the Forward-Backward/Greater Less than 50 task (FB50), the Upper-Lowercase/Divisible or not divisible by 5 task (ULD5), and the Red-Green/X or O (RGXO) task. The FB50 task required alternating between judging whether the letter was facing forward or backward or whether a number was greater or less than 50. The cues used to indicate the task were Forward or Greater than 50 in the upper left hand corner and Backward or Less than 50 in the right hand corner of the screen. In the ULD5 task, participants switched between judging whether a letter was upper or lower case or whether the number was divisible by 5 or not. The corresponding cues were Upper or /5 on the left hand and Lower or Not /5 on the right. In the RGXO task participants were

shown a red or green X or O and alternated between judging the color or the identity of the letter. The cues were Red or X on the left hand and Green or O on the right hand. All stimuli were in 24 point New Courier font.

Procedure. The experiment consisted of two one-hour sessions with the second session occurring 24-48 hours after the first. All participants were tested on random and predictable versions of the CVOE transfer task at the beginning of the first session (*pre-test*) and again at the end of the second session (*post-test*). Pre-test and post-test both consisted of four blocks with 48 trials each: a predictable task-switching block, a random task-switching block, and two single-task blocks in which participants performed only Consonant-Vowel judgments throughout or only Odd-Even judgments. Immediately following pre-test and immediately preceding post-test, all participants performed a 48-trial single-task block of each of the six tasks used during training (Forward or Backward, Greater or Less than 50, Upper or Lower case, Divisible or Not divisible by 5, Red or Green, X or O). This allowed the calculation of mixing costs on the training tasks for the switch-training groups for sessions 1 and 2. Participants were always told the identity of block they would be performing (single-task, random, or predictable).

Training was broken up across sessions with the first 4 blocks of training administered on session 1 and the last four blocks of training on session two. The predictable and random switch-training groups received 8 blocks of practice (4

per day) on each of the three task-switching training tasks for a combined total of 1,152 trials across the three task-pairs. The control group practiced each of the individual tasks alone in twelve single-task blocks of 96 trials across sessions so that the number of trials performed of each individual task was the same between the switch-training and control groups with only the context, task-switching versus single-task, different. Each block was preceded by a practice block of 10 trials on which accuracy feedback was given. These trials were not included in any analyses.

Results

The findings are presented in two sections. In the first we describe the training data. We examined the training data to ascertain that both random and predictable switch training groups showed reliable mixing and switch costs during training and that these costs were reduced with practice. The second section is concerned with the question of transfer of training and whether the training groups showed a greater pre to post-test reduction in cost compared to the control group and each other. To test for mixing and switch costs in both the training and transfer analyses, we specified two contrasts: a mixing cost contrast testing the mean RT of non-switch trials against the mean RT of single-task trials and a switch cost contrast testing the mean RT of switch trials against the mean RT of non-switch trials. For each analysis, we first provide an overview of any main effects and then proceed to the contrasts of interest.

Only RTs from correct trials were used. Any reaction time faster or slower than three standard deviations from the condition mean (single-task, non-switch or switch) for each individual participant were also excluded (0.02% of correct responses). For all analyses in all experiments reported in this paper an alpha of .05 was set.

Training Results. Both the accuracy and RT (see Table 1) data were submitted to a mixed factorial ANOVA with task (FB50, ULD5 and RGXO), trial type (single-task, non-switch and switch), and testing time (first block v. last block) as within-subject factors and group (predictable v. random switch) as a between-subjects factor with two planned comparisons specified; a testing time x switch cost contrast and a testing time x mixing cost contrast.

For the accuracy data, the overall analysis found significant main effects of task (FB50 $M = .90$, ULD5 $M = .98$, RGXO $M = .99$), $F(2, 120) = 1316.9$, partial $\eta^2 = .96$) and trial type (single-task $M = .99$, non-switch $M = .98$, switch $M = .91$), $F(2, 120) = 756.2$, partial $\eta^2 = .93$. Neither contrast was statistically significant, (testing time x switch cost $F(1, 60) = 2.0$, $p = .16$, partial $\eta^2 = .03$ and testing time x mixing cost, $F < 1$) suggesting that any changes seen in mixing or switch cost with practice were not due to any major changes in accuracy with practice.

Insert Table 1 about here

The analysis of the RT data found all main effects to be significant. There was an effect of testing time with an overall reduction in RT from the first to last block, $F(1, 60) = 279.4$, partial $\eta^2 = .82$). There was also an effect of task with larger RTs for the FB50 and ULD5 task and smaller RTs for the RGXO task $F(2, 59) = 199.4$, partial $\eta^2 = .87$). There was a main effect of trial type with larger RTs for switch trials, smaller RTs for non-switch trials and the smallest RTs for the single-task condition, $F(2, 59) = 453.2$, partial $\eta^2 = .94$. Finally there was a main effect of group with overall greater RT for the random training group, $F(1, 60) = 16.2$, partial $\eta^2 = .98$.

Both mixing and switch costs were considerably reduced with practice as both the testing time x mixing cost, $F(1, 108) = 126.6$, partial $\eta^2 = .54$, and the testing time x switch cost, $F(1, 108) = 78.9$, partial $\eta^2 = .42$, comparisons were significant. We ran a follow up analysis adding the factor *group* to the testing time by mixing and testing time by switch comparisons to determine whether there were any differences in the amount of improvement seen in random switch vs. predictable switching. The results of this analysis indicate that there was a trend towards a group difference in switch costs, $F(1, 108) = 3.7$, $p = .058$, partial $\eta^2 = .03$, with larger reduction with practice for the random training group than the predictable training group, but there were not any group differences in the amount of mixing cost reduction, $F < 1$.

Transfer data. Both accuracies and RTs were analyzed using a mixed factorial ANOVA with training group (predictable switch-training, random switch-training or control) as a between-subjects factor and the type of task-switching block (predictable or random), trial type (switch, non-switch or single-task) and testing time (pre-test or post-test) as within-subjects factors. We specified four comparisons to test for evidence of transfer: a group x testing time x switch cost contrast in both the predictable and the random transfer blocks and a group x testing time x mixing cost contrast in both transfer blocks.

The overall accuracy analysis found main effects of transfer type, $F(1, 90) = 5.36$, partial $\eta^2 = .06$) with greater accuracy for predictable switching (predictable $M = .99$ v. random $M = .98$) and trial type, (single-task $M = .98$, non-switch $M = .99$, switch $M = .98$), $F(2, 180) = 6.9$, partial $\eta^2 = .07$. None of the switch or mixing cost comparisons were significant (predictable mixing cost x group x testing time $F(2, 90) = 1.7$, $p = .19$, partial $\eta^2 = .04$, all other F s < 1) suggesting that any pre to post-test reductions in mixing or switch cost were not achieved at the expense of accuracy.

For the RT data, the overall analysis revealed main effects of testing time with all RTs reduced by the second day, $F(1, 90) = 388.4$, partial $\eta^2 = .81$, of trial type with switch RTs longer than non-switch RTs which were larger than the single-task RTs, $F(2, 180) = 914.1$, partial $\eta^2 = .91$, and of the type of switching

block with longer RTs for random switching compared to predictable switching,

$F(1, 90) = 106.0$, partial $\eta^2 = .54$. Figure 1 shows the condition means.

 Insert Figure 1 about here

Of the four comparisons designed to test for transfer of training, only the group x testing time x mixing cost comparison in the random transfer block was significant, $F(2, 90) = 6.9$, partial $\eta^2 = .13$. While a visual inspection of the mixing costs for the predictable transfer condition, shown in Figure 2a, suggests a greater pre to post-test reduction for the predictable training group, this trend was not statistically significant, $F(2, 90) = 1.2$, $p = .30$, partial $\eta^2 = .02$. In addition, neither switch cost comparison was significant, $F_s < 1$.

 Insert Figure 2 about here

To determine the exact nature of improvement in the random transfer block as measured by mixing cost, we tested each training group separately against the control group and against each other. The random switch training group showed significantly greater pre to post-test reductions in mixing cost for the random transfer block against both the control group, $F(1, 60) = 11.5$, partial $\eta^2 = .16$, and the predictable switch training group, $F(1, 60) = 9.7$, partial $\eta^2 = .14$,

while the predictable training group and the control group did not differ from each other, $F < 1$.

Discussion

This experiment was designed to examine whether practice-based improvements frequently observed in task-switching performance can generalize to a different switching context. While both predictable and random switch-training groups demonstrated clear practice-based reductions in both mixing cost and switch cost for the three different pairs of tasks trained on, strong evidence of transfer was only observed from practice on random switching to the random switching block of the transfer task as measured by mixing cost. Switch cost on the other hand showed no clear pattern of transfer in either the random or the predictable transfer blocks. These results present a mixed picture of the extent to which transfer of training is possible in the task-switching paradigm. Compared to a control group that did not practice switching between tasks, the random switch-training group showed greater reductions in both switch and non-switch RTs on a randomly switching transfer task. However, this superior performance did not appear to transfer to a predictable block of the same transfer task. There was a trend for a similar pattern for the predictable training group with a greater pre to post-test reduction of mixing cost in the predictable block of the transfer task although this did not reach statistical significance.

The lack of transfer between the two types of switching suggest that the benefits observed are unlikely to be due to simple increases in arousal after practicing switching. It would also appear to argue against general improvements in the resolution of stimulus ambiguity or inhibiting an inappropriate response since these presumably would be common to both types of switching. Earlier we noted that differences in cost have been reported between random and predictable switching. This difference is largely due to the non-switch trials. Rogers & Monsell (1995) initially argued that task-set reconfiguration is complete with the first non-switch trial and that no further adjustments are necessary. In their data which used predictable switching, subsequent non-switch trials showed no further reductions in RT. However, later studies using random switching reported a gradual decline in RT across non-switch trials (Meiran, Chorev & Sapir, 2000; Monsell et al., 2003, Salthouse, Fristoe, McGuthry & Hambrick, 1998). Milán, Sanabria, Tornay, & González (2005) interpreted this difference in the pattern of performance following a switch as evidence that different mechanisms are necessary in reconfiguring the task-set for predictable and for random switching. An alternate view put forth by Monsell et al. (2003) is that the same mechanism is used to reconfigure task-set in both predictable and random switches. However, when the identity of future trials is uncertain as in random switching, the task-set bias may be attenuated as participants do not fully commit to current task-set.

Either interpretation may help explain the apparent lack of transferable improvement between random and predictable switching. It is worth noting that in our data the transferable improvement seen affected both the switch and non-switch trials roughly equally so that the difference between the training groups was reflected in mixing cost not switch cost whereas hypotheses concerning task-set reconfiguration are concerned with the difference between the two trial types.

Experiment 2

In this experiment, we attempted to replicate the transfer of training seen in Experiment 1. We used the same procedure with one modification, the addition of a short interval of 200 milliseconds (ms) between the cues indicating the task to be performed and the target stimulus. This change served two purposes. First, a replication of our earlier result using a different cue-to-stimulus interval (CSI) would help establish that our earlier finding was not peculiar to rapid switching with no opportunity for preparation. Second, the addition of a preparatory interval between the cue and stimulus may ameliorate possible differences in task-set bias between predictable and random switches i.e. if participants know they will have a heads up on the next trial type, they may show a greater commitment to the current task set. This may allow for transfer from training on one type of switching to another.

Method

Participants. Seventy-five University of Michigan undergraduates participated for a \$20 payment and were randomly assigned to the random switch-training group ($N = 25$), the predictable switch-training group ($N = 25$) and the control group ($N = 25$).

Procedure. The design, tasks and procedure were the same as those reported in Experiment One with one change. A 200 ms CSI was used for all switching blocks, both random and predictable, throughout the experiment.

Results

The data were analysed exactly as in Experiment 1.

Training data. Data from two participants were lost due to computer error and are therefore excluded from the analyses of training data.

The overall analysis of training accuracy found main effects of task (FB50 $M = .91$, ULD5 $M = .98$, RGXO $M = .98$), $F(2, 92) = 179.2$, partial $\eta^2 = .79$, trial type (single-task $M = .98$, non-switch $M = .97$, switch $M = .92$), $F(2, 92) = 262.2$, partial $\eta^2 = .85$ and group (random $M = .96$, predictable $M = .95$), $F(1, 46) = 4.24$, partial $\eta^2 = .08$. Neither contrast was statistically significant, testing time x switch cost, $F(1, 46) = 2.6$, $p = .11$, partial $\eta^2 = .05$, testing time x mixing cost, $F < 1$, suggesting no significant change in accuracy from the first block of training to the last.

For the RT data (see Table 2), the results were similar to Experiment 1, there were main effects of testing time, $F(1, 46) = 310.3$, partial $\eta^2 = .87$, task,

$F(2, 45) = 220.7$, partial $\eta^2 = .83$, and trial type, $F(2, 45) = 324.9$, partial $\eta^2 = .88$.

Specifically, RTs decreased significantly from the first block of training to the last block, the slowest RTs were for the FB50 task and the fastest for the RGXO and the longest RTs were for switch trials and the shortest for single-task trials.

However, unlike Experiment 1, there was no overall effect of group, $F(1, 46) = 1.1$, $p = .30$, partial $\eta^2 = .02$.

Insert Table 2 about here

Transfer data. The overall accuracy analysis found main effects of transfer type, $F(1, 72) = 12.3$, partial $\eta^2 = .15$, with better accuracy for predictable switching ($M = .98$ v. random $M = .97$), and trial type (single-task $M = .98$, non-switch $M = .98$, switch $M = .97$), $F(1, 72) = 42.5$, partial $\eta^2 = .37$. None of the contrasts were significant, predictable: testing time x group x switch $F(2, 72) = 1.5$, $p = .23$, partial $\eta^2 = .04$, all other F s < 1 .

In the overall analysis of the RT data (shown in Figure 3), there were main effects of testing time, $F(1, 72) = 205.8$, partial $\eta^2 = .74$, trial type, $F(1, 72) = 1206.3$, partial $\eta^2 = .94$ and transfer type, $F(1, 72) = 54.0$ partial $\eta^2 = .43$, similar to those seen in Experiment 1.

Insert Figure 3 about here

The contrast results mirrored those seen in Experiment 1 with only the group x testing time x mixing cost comparison for the random switching transfer block reaching significance, $F(2, 72) = 6.9$, partial $\eta^2 = .16$. Testing each training group separately against the control group and against each other found the random training group had significantly greater pre to post-test reductions in mixing cost for the random transfer block against both the control group, $F(1, 48) = 16.9$, partial $\eta^2 = .26$, and the predictable switch training group, $F(1, 48) = 4.9$, partial $\eta^2 = .09$, while the predictable training group and the control group did not differ from each other, $F(1, 48) = 1.6$, $p = .21$, partial $\eta^2 = .03$.

The predictable mixing cost comparison was not significant, $F(2, 72) = 1.6$, $p = .20$, partial $\eta^2 = .04$. As in Experiment 1, a visual inspection of the data hinted of a difference between at least one of the training groups and the control group at post-test. However, this time it was the random training group with a trend towards a greater reduction in mixing cost compared to the control group, $F(1, 48) = 3.18$, $p = .08$, partial $\eta^2 = .06$, while the predictable group did not differ from the control group, $F < 1$. Neither of the switch comparisons were significant, predictable testing time x group x switch cost, $F(2, 72) = 1.4$, $p = .25$, partial $\eta^2 = .04$ and random testing time x group x switch cost, $F(2, 72) = 1.5$, $p = .24$, partial $\eta^2 = .04$.

Insert Figure 4 about here

Discussion

This experiment attempted to replicate the findings of Experiment 1 with the addition of a short preparatory interval of 200 ms between the cue and the target stimulus. The pattern of results seen was very similar. The only statistically reliable evidence of transferable improvement was for the random training group for the random transfer mixing cost comparison. However, as in Experiment 1, a visual examination of the predictable transfer data suggested a greater pre to post-test reduction for at least one of the training groups. In the first study, the predictable training group appeared to have a slight advantage, but in Experiment 2, it was the random switch-training group that appeared to show greater improvement compared to the control group. In neither experiment, did these differences reach statistical significance nor did any pre to post-test differences in switch cost. To test whether these differences could become significant with a larger N, we combined the data across Experiments 1 and 2, ran the same contrasts and found the same results with a robust pre-test to post-test mixing cost improvement for the random training group on the random transfer task, $F(2, 165) = 13.6$, partial $\eta^2 = .14$, a non-significant difference in the predictable mixing contrast, $F(2, 165) = 1.4$, $p = .24$, partial $\eta^2 = .02$, and no reliable differences in switch costs, $F_s < 1$.

The results from these experiments raise two main questions; 1) what is it that is improving with the random transfer and 2) why are there significant effects for random switching, but not predictable? As we noted earlier, random and predictable switching involve many of the same processes. What differs is knowledge of the identity of the next trial type and the necessity of the cues. In predictable switching, if participants are keeping track of the sequence then they know with 100% certainty whether the next trial will continue the present task or will switch. However, in random switching, the identity of the next trial is uncertain until the next cue appears. This level of uncertainty in random switching has been hypothesized as leading participants to commit less fully to the current task set (Monsell et al., 2003). Dreisbach and colleagues have also hypothesized that the level of cognitive control exerted can be adjusted based on the probability of a switch occurring. Their work using cues that predict the next trial with varying degrees of certainty has shown that expectancies affect performance on both switch and non-switch trials (Dreisbach & Haider, 2006; Dreisbach et al., 2002). Therefore, what participants may be learning over training and then transferring to new task pairs is an optimal adjustment of task-set bias on different trial types. This may explain the greater degree of transfer for random switching. In predictable switching, the set of expectancies can be easily learned while in random switching, participants likely need greater amounts of practice. There is some evidence for this in the training data. While there was no interaction

between testing time and training group when comparing the first and the last blocks of training, when we tested the first block of training v. the fifth block of training (i.e. the first block of training on the second day of the study), there was a significant interaction between testing time and group with the predictable switching group showing a larger decrease in mixing cost, $F(2, 216) = 17.9$, partial $\eta^2 = .14$. However, this difference in improvement was gone after three additional blocks of practice. Therefore, one reason the post-test differences between groups on the predictable transfer task were so weak was that the control group may have some appreciable improvement from their pre-test exposure to predictable switching.

A second reason that predictable transfer was so weak may be that participants are using different strategies in switching predictably. In our variant of the task in which each trial is cued as well as occurring in a predictable sequence, participants may use either their knowledge of sequence or rely on the cues. Therefore, it is possible that results seen in the predictable transfer data are due to mixture of different strategies across participants. We included cues in our predictable switching so that working memory demands would not differ between predictable and random. In addition, cues allowed participants to get back on sequence if they made an error. Without cues, it is easy for a participant to lose track of which trial they are performing and then to be off-sequence for the rest of a block even though after the mistake participants may think they are performing

the task accurately. This can make error analysis difficult. Unfortunately, while trying to avoid these problems, we may have inadvertently created another. Future research using predictable switching should either include measures such as monitoring eye movements to determine whether participants are relying on the cues or use cueless predictable switching.

In random switching, participants must rely upon the cues to give them the identity of the current task. Therefore, a second possible source of improvement with practice is the use of the cues. During the training component, participants may have improved in retrieving a particular task set in response to a specific cue. Naturally, this would not account for any improvement in switching between a new pair of tasks. However, it may be that participants are learning a strategy specific to the type of cue used during both training and transfer, verbal task labels placed in the corners of the screen. Given that the cues used in Experiments 1 and 2 appear in corners of screen, participants may have learned to detect changes in the cues without having to saccade to the corners. This would have improved RT on both switch and non-switch trials. Therefore, in our last experiment, we tested whether we could replicate the pattern of transfer seen from random training to a random post-test measure using different types of cues at training and at pre and post-test.

Experiment 3

In Experiment 3, we tested whether the transferable improvement on random switching with practice may be seen across different cuing paradigms. In random switching, participants are entirely dependent on external, explicit cues, and these can vary in the degree of their transparency. For example, in Experiments 1 and 2, the words “Consonant or Vowel” or “Odd or Even” were used to indicate which task should be performed. These cues are highly transparent. However, many experiments use more arbitrary cues such as the color of a box surrounding the stimulus or the location of the stimulus to indicate which task should be performed. In our first two experiments, the same type of verbal and highly transparent cues was used during both the training and transfer portions of the experiment. This raised the possibility that any transfer observed may have been due to participants learning cue-specific strategies.

However, if the transfer is seen from one type of cue to another, then any improvements in switching performance at post-test are unlikely to be due to a cue-specific strategy. Therefore, we used two cues types at pre and post-test, the highly transparent verbal cue used in our earlier experiments and a second more arbitrary cue based on the spatial location of the stimulus. Participants in the switch-training group received practice on the same three training tasks used in our earlier experiments, but using both the spatial cue used at pre-test and a second type of arbitrary cue, the color of a box surrounding the target stimulus, rather than verbal cues. A control group practiced the same tasks in a non-

switching context. If transferable improvement is the result of particular demands or strategies associated with a specific type of cue, we should expect transfer only for the spatial cue used during both training and pre and post-tests. However, if transfer results from changes in processes that are independent of the cues practiced during training, we should expect to see transferable improvement for both the verbal and the spatial cue.

Method

Participants. 48 University of Michigan and Washington University students participated for a payment of \$20 or course credit. Participants were randomly assigned to the switch-training ($N = 24$) or control ($N = 24$) group.

Design. We employed a pre-test/post-test design similar to Experiments 1 and 2 with three major changes. The first was that we had only two groups, a random switch-training group and a control group which practiced single-task performance during the training portion of the study. In addition, given that we no longer had to match the number of switch and non-switch trials to a predictable condition, we allowed tasks to switch entirely randomly. The final change was the addition of a new within-subjects variable, the type of cue. During switch-training, there were two types of cue indicating which task was to be performed, a line cue and a box cue (described below) and at transfer, there was a switching block which used the line cue and one which used the same verbal cues as used in Experiments 1 and 2.

Tasks and Stimuli. The training and transfer tasks were the same as those in Experiments 1 and 2. In all switching blocks, tasks switched unpredictably with a CSI of 200 ms.

The Consonant-Vowel/Odd-Even task was again used to measure transfer. However, this time two types of cues were used to indicate the task to be performed. One was the explicit verbal cue used earlier with the words *Consonant* and *Vowel* presented in the corners of the screen when the letter judgment was required and the words *Odd* and *Even* when the number judgment was required. The second was the line cue where the letter-number pair appeared either above or below a black line in the center of the screen. If the pair was above the line, participants made the letter judgment, and if it was below the line, they made the number judgment.

The same three task-switching training tasks were practiced by the random switch-training group. Two types of cues were used during training: the spatial line cue used in one transfer block and a box cue. In the box cue, the target stimulus appeared in a box located in the center of the screen. If the box was blue then participants performed one task and if was yellow, they performed the other task. A card with all the task mappings was placed next to the computer in case a participant forgot which task to perform if the stimulus was above or below the line or in a blue or yellow box on any particular task. All stimuli were presented in 24 point New Courier font. The verbal cue was never used during training.

Procedure. The experiment consisted of two one-hour sessions with the second session occurring 24-48 hours after the first. Both groups of participants were measured on the Consonant-Vowel/Odd-Even transfer task at the beginning of the first session (*pre-test*) and again at the end of the second session (*post-test*). At both pre-test and post-test, participants first completed two single-task blocks, a 48 trial block of solely Consonant-Vowel judgments and a 48 trial block of Odd-Even judgments. Both the verbal and line cues were present for the single-task block. Participants then completed two Consonant-Vowel/Odd-Even switching blocks of 64 trials each with one block using the line and the other using the verbal cue. The order of these blocks was counterbalanced across participants. The remainder of session 1 and the beginning of session 2 consisted of the training component of the study. Both switch-training and control groups first completed a 48 trial single-task block of each of the six component tasks that made up the three switching tasks with the line cue present on the first 24 trials and the box cue present on the second 24 trials. The switch-training group then practiced eight 48 trial blocks of each of the three switching tasks with half the blocks using the line cue and half using the box cue for a combined total of 1,152 mixed-task trials across both days of training. As in Experiments 1 & 2, the control group practiced each of the individual tasks alone in twelve single-task blocks of 96 trials. The same cues and spatial location of the stimulus were used for the single-task blocks as for the switching blocks so that the switch-training

and the control group received the exact same visual display on any given trial.

Each block was preceded by a practice block of 10 trials on which accuracy feedback was given. These trials were not included in any analyses.

Results

Training Results. Data from one participant were lost due to computer error. Both the accuracy and RT data were submitted to a within-subjects ANOVA with testing time (first block v. last block), task (FB50, ULD5 and RGXO) trial type (single-task, non switch and switch) and cue type (line and box) as factors. Practice effects were examined by specifying two contrasts, testing time x mixing cost and testing time x switch cost, for the two different cue types.

In the overall accuracy analysis, there were main effects of task (FB50 $M = .95$, ULD5 $M = .96$, RGXO $M = .95$), $F(2, 44) = 3.6$, partial $\eta^2 = .14$, cue type (Line $M = .96$, Box $M = .95$), $F(1, 22) = 4.3$, partial $\eta^2 = .16$, and of trial type (single-task $M = .97$, non-switch $M = .96$, switch $M = .95$), $F(2, 44) = 10.1$, partial $\eta^2 = .32$, with greater accuracy for the single-task trials. However, none of the practice based contrasts were significant (Box cue: testing time x switch $F(1, 22) = 1.9$, $p = .18$, partial $\eta^2 = .08$, all other F s < 1).

For the RT data (see Table 3), there were main effects of testing time with a reduction in overall RT from the first blocks to the last, $F(1, 22) = 114.8$, partial $\eta^2 = .84$), of trial type, $F(2, 21) = 180.8$, partial $\eta^2 = .95$, of task, $F(2, 21) = 27.6$, partial $\eta^2 = .72$, with the longest RTs for the FB50 task, and of cue type, $F(1, 22)$

= 195.7, partial $\eta^2 = .90$, with longer RTs for blocks using the box cue. Both switch cost contrasts were significant indicating robust practice effects for switch cost for both line, $F(1, 22) = 38.74$, partial $\eta^2 = .64$, and box, $F(1, 22) = 43.57$, partial $\eta^2 = .66$, cue types. There was a significant reduction in mixing cost in the line cue condition, $F(1, 22) = 7.51$, partial $\eta^2 = .25$. There was also a reduction in mixing cost from the first to the last block of training for the box cue condition though this did not reach the level of statistical significance set for this study, $F(1, 22) = 3.9$, $p = .06$, partial $\eta^2 = .15$.

Insert Table 3 about here

Transfer Results.

The accuracy data were submitted to a repeated measures ANOVA with training group (random training v. control) as a between-subjects factor and the type of cue (line v. verbal), trial type (switch, non-switch or single-task) and testing time (pre-test v post-test) as within-subjects factors. Four contrasts were specified: a group x testing time x mixing cost and a group x testing time x switch cost comparison in both the line cue transfer block and the verbal cue transfer block.

Accuracy was overall quite high. The overall analysis found only a main effect of trial type (single-task $M = .97$, non-switch $M = .97$, switch $M = .95$), $F(1,$

46) = 18.7, partial $\eta^2 = .30$. None of the mixing or switch contrasts were significant, verbal: group x testing time x mixing cost, $F(1, 46) = 1.2, p = .28$, partial $\eta^2 = .03$ with all other F s < 1.

The same analysis was applied to the RT data (see Figure 5). All main effects were significant. There an overall decline in RT from pre-test to post-test, $F(1, 46) = 421.9$, partial $\eta^2 = .90$. RTs were affected by the type of cue used with faster RTs for the line cue, $F(1, 46) = 210.1$, partial $\eta^2 = .82$, as well as the trial type with longer RTs on switch trials, $F(2, 92) = 474.1$, partial $\eta^2 = .91$. Finally, there was a larger pre to post-test reduction in RT for the training group compared to the control group, $F(1, 46) = 9.7$, partial $\eta^2 = .17$. However, no interactions involving testing time, group and cue type were significant, F s < 1, indicating that any pre to post-test differences between the training and control groups were not due to the type of cue used.

 Insert Figure 5 about here

Both the line and verbal cue mixing cost comparisons were significant, Line $F(1, 46) = 12.16$, partial $\eta^2 = .21$, Verbal $F(1, 46) = 4.4$, partial $\eta^2 = .09$, with a greater pre to post-test reduction in mixing cost for the switch-training group compared to the control group as shown in Figure 6. Neither switch cost contrast

was significant, Line $F(1, 46) = 1.2, p = .29$, partial $\eta^2 = .03$, Verbal $F(1, 46) = 2.5, p = .12$, partial $\eta^2 = .05$.

 Insert Figure 6 about here

Discussion

The results of this experiment demonstrate that the transferable improvement seen after training on random task switching can occur between different types of cues. The training group showed larger reductions in the mixing cost comparison than the control group for both the line cue practiced during training and for the verbal cue that was not practiced. The pre to post-test reductions on the verbal cue transfer task for the training group in this experiment (223 ms) were similar in magnitude to those seen in Experiment 2 (250 ms). This indicates that the transfer observed in all three experiments was not the result of cue specific strategies.

The design of Experiment 3 also made it unlikely that the training group learned to better anticipate when switch and non-switch trials would occur. This was a possibility in both Experiments 1 and 2 where random switching was constrained so the number of switch and non-switch trials in a block would match the predictable switching condition. With practice, participants may have learned to implicitly gauge the likelihood of a switch or non-switch based on previous

trials. However, in Experiment 3, trials were allowed to switch completely at random. As a result, sequences where two tasks alternated rapidly in succession or where 3 or more non-switch trials occurred in a row were not unusual and had no effect on the trial composition of the rest of the block.

A third possibility was that training on random switching improved the participants' ability to resolve task-set competition. Task-set competition, theoretically, should be highest on trials following a switch since both task sets will have been active on the previous two trials. Therefore, we decided to run a secondary analysis in which we not only considered whether the current trial was a switch or a non-switch, but also the identity of the trial immediately preceding it. To do so, we collapsed across cue types and rebinned the trials into four new trial types, a switch following a switch (SwSw), a switch following a non-switch (NsSw), a non-switch following a switch (SwNs) and a non-switch following a non-switch (NsNs). We submitted these RTs to a repeated measures ANOVA with training group as a between-subjects factor and testing time, current trial type and preceding trial type as within-subjects factors. There was a significant interaction between testing time, group and the preceding trial type with greater pre to post-test improvements for the training group on trials preceded by a switch, $F(1, 46) = 6.8$, partial $\eta^2 = .13$. The four-way interaction with current trial type did not reach significance although a trend was present $F(1, 46) = 3.5$, $p =$

.07, partial $\eta^2 = .07$. However, this was difficult to interpret given the pre-test differences between groups for the SwSw trials.

In light of the 3-way interaction between testing time, group and preceding trial type, we collapsed over the current trial type and tested the pre to post-test improvement seen for the training group for trials following a switch against the training group improvement for trials following a non-switch, $F(1, 23) = 13.8$, partial $\eta^2 = .37$, against the control group improvement for trials following a switch, $F(1, 46) = 23.2$, partial $\eta^2 = .34$, and trials following a non-switch, $F(1, 46) = 31.9$, partial $\eta^2 = .41$. There was no pre to post-test difference between trials following a switch and those following a non-switch for the control group, $F < 1$.

 Insert Figure 7 about here

As seen in Figure 7, the transfer effect is strongest for trials that immediately follow a switch and weaker for trials following a non-switch. Therefore, it appears that the training group is faster at recovering *from an unexpected switch*, and this improved performance on the following trial whether it was a switch or non-switch.

General Discussion

Evidence of transferable improvement in task-switching performance is of considerable interest for our understanding of both task-switching and the

possibility of transferable improvement in executive functions. The most consistent evidence of transfer across three experiments was seen for random switching as measured by the mixing cost comparison. In contrast, there was little evidence of transfer as measured by switch cost. The reason for this became apparent in Experiment 3 where we learned that transfer was the strongest for trials immediately following a switch, and this was true whether the trial was another switch trial or a non-switch.

Thus, what is improving with training can best be characterized as recovering from an unexpected switch, unexpected because we found no evidence of transfer from practice on predictable switching to random. As discussed above, one well-known difference between predictable and random switching lies in the first non-switch trial (Monsell et al., 2003). In predictable switching, the first non-switch trial does not differ from ensuing non-switch trials while in random switching, participants need several non-switch trials to fully recover from a switch. This has been attributed to differences in task set bias in random switching with participants unwilling to commit fully to the new task set. However, the data from our final experiment argue against any deliberate biasing of task set by participants since *both* switch and non-switch RTs are larger following a switch trial, and both show improvement at transfer. Therefore the difference between random and predictable switching may be that a switch is less disruptive if it is expected. This is similar to recent work looking at task interruptions in which

participants who were warned of an interruption recovered more quickly than participants who were not given warning (Trafton, Altmann, Brock & Mintz, 2003). Especially intriguing from our perspective was that with practice, participants who received no warning improved their ability to recover from a task interruption with practice.

A number of different processes have been hypothesized as being involved in switching tasks. Based on these initial data, we propose that the most likely source of transferable improvement lies in the training participants' ability to resolve task set conflict (Kray & Lindenberger, 2000; Mayr, 2003). Task set competition would be especially large following a switch since both task sets would have been active in the two previous trials. The successful resolution of such competition may be accomplished through increased attentional control (Hübner, Futterer, & Steinhauser, 2001), additional set-selection or decision processes (Rubin & Meiran, 2005; Steinhauser & Hübner, 2005) or inhibitory processing at the level of either the task set, the response set or the stimulus (Mayr, 2001).

Of these three, inhibition would appear to be the least likely prospect as there is evidence that task-set inhibition is automatic and not penetrable by preparation (Dreisbach et al., 2002; Mayr & Keele, 2000). Stimulus and response based inhibition also appear unlikely. As we noted earlier, the lack of transfer between predictable and random switching would argue against stimulus and

response based processes. Instead improvements in attentional control represent a more likely prospect. A sequence effect similar to the one seen in Experiment 3 has been observed in an attention switching task in which participants had to maintain two internal counters (Gehring et al., 2003). A task such as this would be an excellent candidate for a future test of transfer. If training on task switching leads to improved conflict resolution between two internal representations, then transfer from task switching to an attention switching task such as the one described above could be possible. However, if the transfer seen in task-switching is the result of improvements in decision processes specific to task sets, then transfer would not occur.

Task switching performance as measured by switch cost did not appear to show much evidence for transferable improvement in the task switching paradigms used in these experiments. A lack of transfer is consistent with some theoretical interpretations of the switch cost such as those attributing switch costs to task or stimulus level priming (e.g., Wylie & Allport, 2000). The pattern of results seen here naturally can not rule out the possibility of transferable improvements in switch costs in future studies using different variations of the task-switching paradigm. However, for researchers interested in training or remediation of executive function, these results illustrate that within a given executive task, there may be multiple measures of a putative executive process and that one may be more sensitive or appropriate for the measure of transferable

improvement than another. Many experiments investigating task-switching have only reported switch cost as their dependent measure. In terms of our study, this would have resulted in no evidence of transferable improvement in task-switching.

Conclusion

The study of transferable improvement in task-switching not only adds to our understanding of the different processes involved in task-switching, but also adds to a small but growing literature of studies demonstrating transferable improvement in executive processes (Bherer et al., 2005; Dowsett & Livesey, 2000; Klingberg et al., 2002; Kramer et al, 1995). Given the rising interest in the extent to which executive processes such as task-switching can benefit from various interventions such cognitive training or rehabilitation programs in impaired populations (Butler & Copeland, 2002; Kramer, Hahn, & McAuley, 2000; Minear & Shah, 2006; Rueda, Rothbart, McCandliss, Saccomano, & Posner, 2005), it is important to establish the extent and the conditions under which training on these processes can yield measurable improvements.

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Table 1

Experiment 1 Mean RTs (ms), mixing and switch costs (ms difference) with standard deviations (in parentheses) for the first and last blocks of the three training tasks, the Forward-Backward/Greater-Less than 50 (FB50) task, Upper-Lower/Divisible by 5 (ULD5) and the Red-Green/X or O (RGXO) task

		Predictable Training Group					Random Training Group				
		Single	Non-	Switch	Mixing	Switch	Single	Non-	Switch	Mixing	Switch
		Task	switch	Trials	Cost	Cost	Task	Switch	Trials	Cost	Cost
		Trials					Trials				
FB50	First Block	751	1020	1395	269	375	794	1271	1691	477	420
		(109)	(323)	(309)	(240)	(177)	(182)	(271)	(306)	(179)	(190)
	Last Block	685	740	1011	55	271	707	1026	1412	319	386
		(102)	(169)	(245)	(128)	(152)	(124)	(215)	(313)	(151)	(187)
ULD5	First Block	647	926	1333	279	407	677	1218	1619	541	401
		(90)	(282)	(294)	(205)	(180)	(113)	(229)	(208)	(192)	(205)

RGXO	Last Block	589	714	973	125	259	603	986	1276	383	290
		(64)	(130)	(199)	(93)	(120)	(91)	(178)	(207)	(140)	(139)
	First Block	588	889	1027	301	138	616	913	1022	297	109
		(95)	(183)	(180)	(146)	(145)	(102)	(161)	(180)	(118)	(109)
	Last Block	552	697	763	145	66	595	702	735	107	33
		(76)	(119)	(134)	(96)	(157)	(113)	(166)	(157)	(101)	(44)

Table 2

Experiment 2 Mean RTs (ms), mixing and switch costs (ms difference) with standard deviations (in parentheses) for the first and last blocks of the three training tasks, the Forward-Backward/Greater-Less than 50 (FB50) task, Upper-Lower/Divisible by 5 (ULD5) and the Red-Green/X or O (RGXO) task

		Predictable Training Group					Random Training Group				
		Single	Non-	Switch	Mixing	Switch	Single	Non-	Switch	Mixing	Switch
		Task	switch	Trials	Cost	Cost	Task	Switch	Trials	Cost	Cost
		Trials					Trials				
FB50	First Block	739	975	1403	236	428	697	891	1256	194	365
		(94)	(239)	(289)	(210)	(222)	(154)	(169)	(187)	(173)	(167)
	Last Block	642	672	929	30	257	658	702	966	44	264
		(82)	(170)	(251)	(141)	(157)	(121)	(106)	(182)	(118)	(157)
ULD5	First Block	645	846	1256	201	410	654	806	1160	152	354
		(68)	(236)	(237)	(209)	(182)	(179)	(189)	(248)	(205)	(167)

	Last Block	574	654	817	80	163	557	656	832	99	176
		(71)	(176)	(208)	(170)	(81)	(71)	(124)	(144)	(105)	(87)
RGXO	First Block	540	675	810	135	135	535	657	717	122	60
		(63)	(134)	(125)	(122)	(94)	(172)	(133)	(165)	(193)	(72)
	Last Block	511	522	575	11	53	482	486	510	4	24
		(65)	(66)	(79)	(43)	(46)	(71)	(87)	(104)	(70)	(53)

Table 3

Experiment 3 Mean RTs (ms), mixing cost, switch cost (ms difference) and standard deviations for the first and last blocks of the three training tasks, the Forward-Backward/Greater-Less than 50 (FB50) task, Upper-Lower/Divisible by 5 (ULD5) and the Red-Green/X or O (RGXO) task

		Line Cue					Box Cue				
		Single	Non-	Switch	Mixing	Switch	Single	Non-	Switch	Mixing	Switch
		Task	switch	Trials	Cost	Cost	Task	Switch	Trials	Cost	Cost
		Trials					Trials				
FB50	First Block	683	673	992	-10	319	677	807	1194	130	387
		(83)	(91)	(239)	(75)	(192)	(73)	(171)	(187)	(166)	(160)
	Last Block	626	578	722	-48	144	621	678	1001	57	323
		(100)	(94)	(153)	(109)	(202)	(99)	(121)	(169)	(136)	(113)
ULD5	First Block	577	596	795	19	199	569	726	1145	157	419
		(48)	(88)	(156)	(166)	(114)	(47)	(175)	(296)	(154)	(267)
	Last Block	524	501	622	-23	121	502	727	874	225	121

		(53)	(55)	(103)	(66)	(77)	(48)	(104)	(119)	(75)	80
RGXO	First Block	450	600	901	150	301	431	742	1148	311	406
		(38)	(127)	(125)	(124)	(92)	(41)	(179)	(194)	(200)	(182)
	Last Block	410	545	727	135	183	399	608	920	275	318
		(35)	(108)	(104)	(103)	(71)	(37)	(124)	(160)	(90)	(105)

Figure Captions

Figure 1. Figure 1a shows the RT means for the predictable transfer task at pre-test and post-test broken down by trial type (switch, non-switch and single-task) and group (random switch-training, control and predictable switch-training) in Experiment 1. Figure 1b shows the same RT means for the random transfer task. Error bars reflect standard error for all figures.

Figure 2. Figure 2a shows mixing and switch costs for predictable transfer in Experiment 1. Figure 2b shows mixing and switch costs for random transfer in Experiment 1.

Figure 3. Figure 3a shows the RT means for the predictable transfer task at pre-test and post-test broken down by trial type (switch, non-switch and single-task) and group (random switch-training, control and predictable switch-training) in Experiment 2. Figure 3b shows the same RT means for the random transfer task.

Figure 4. Figure 4a shows mixing and switch costs for predictable transfer in Experiment 2. Figure 4b shows mixing and switch costs for random transfer in Experiment 2.

Figure 5. Figure 5a shows the individual mean RTs broken down by trial type and group for the line version of the transfer task in Experiment 3. Figure 5b shows the same data for the verbal transfer task in Experiment 3.

Figure 6. Figure 6a shows the mixing and switch cost for the line transfer in Experiment 3. Figure 6b shows the mixing and switch cost in the verbal transfer block in Experiment 3.

Figure 7. Figure 7a shows the mean RTs for Switch trials as a function of testing time, group and the identity of the previous trial in Experiment 3 while Figure 7b shows the same information for the Non-switch trials.

List of Figures

Figure 1a

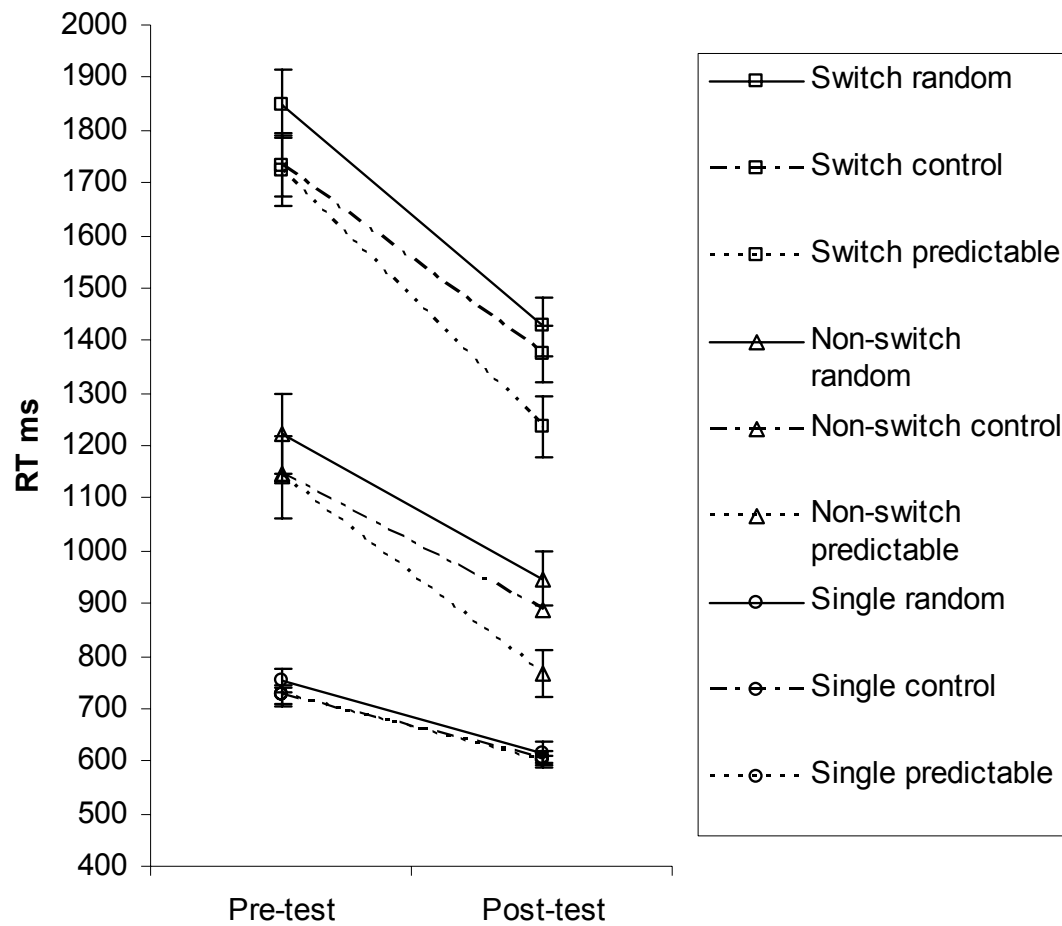


Figure 1b

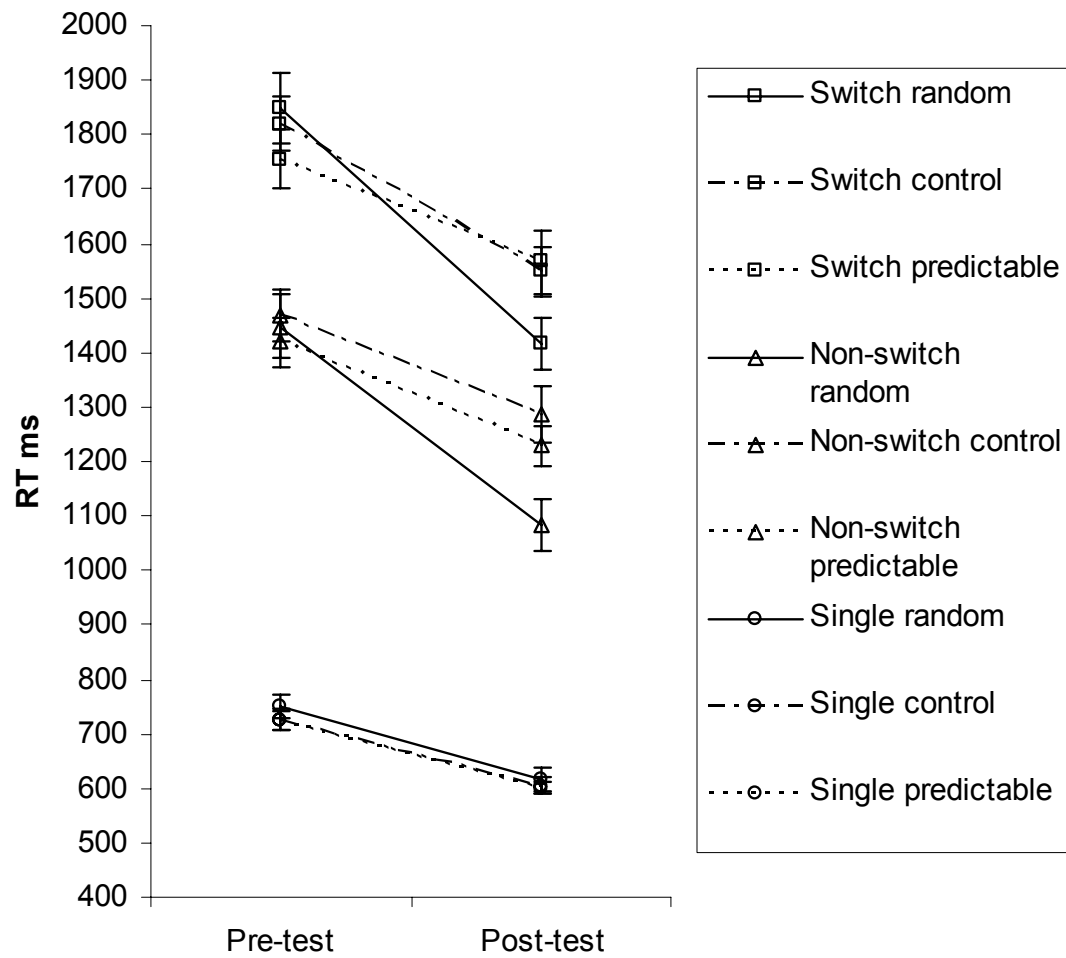


Figure 2a

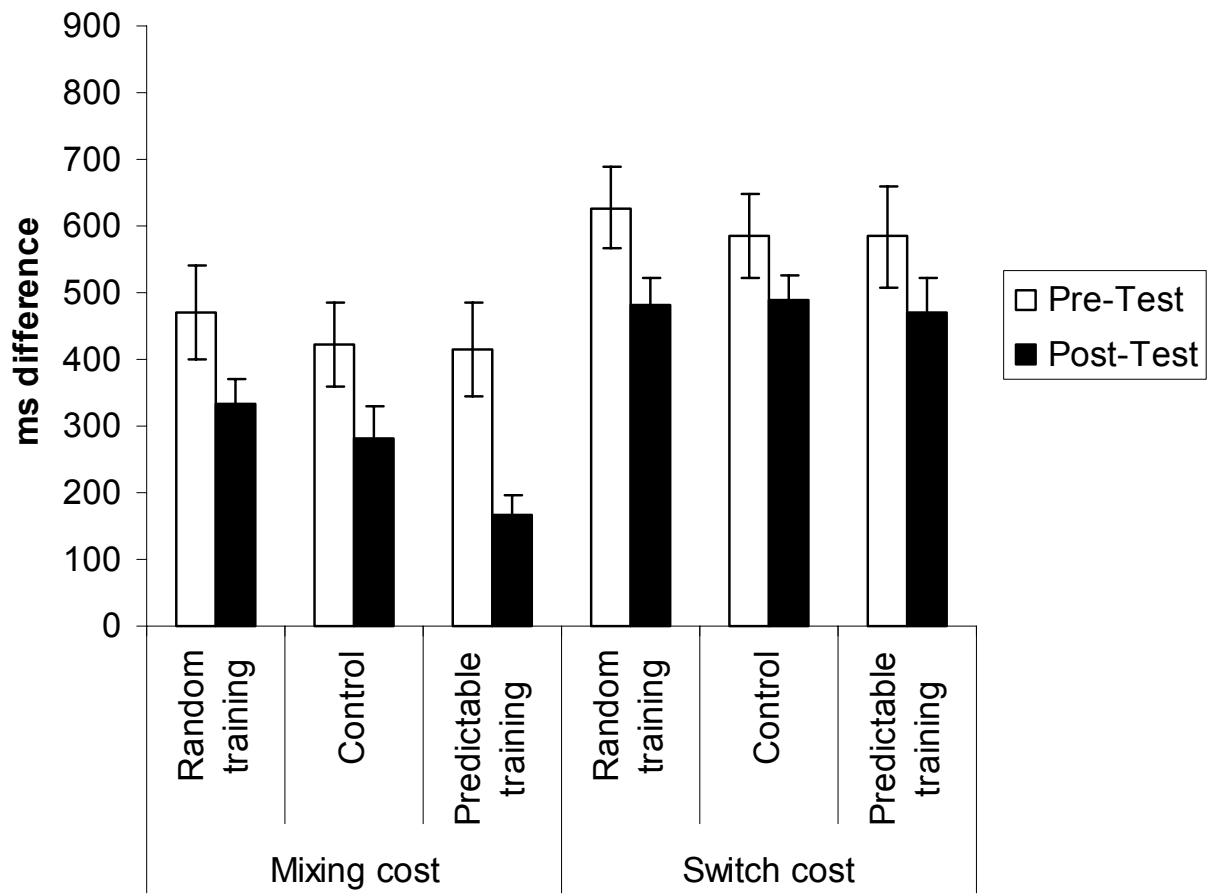


Figure 2b

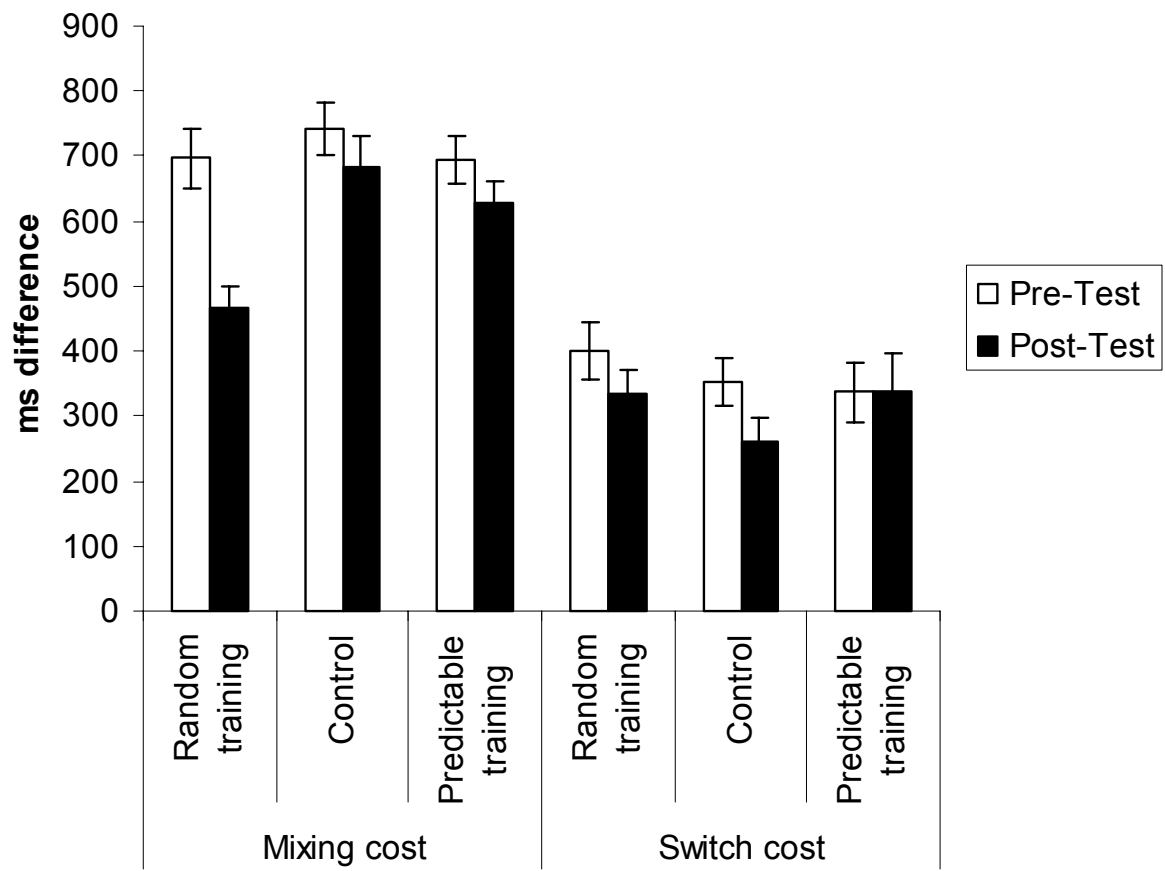


Figure 3a

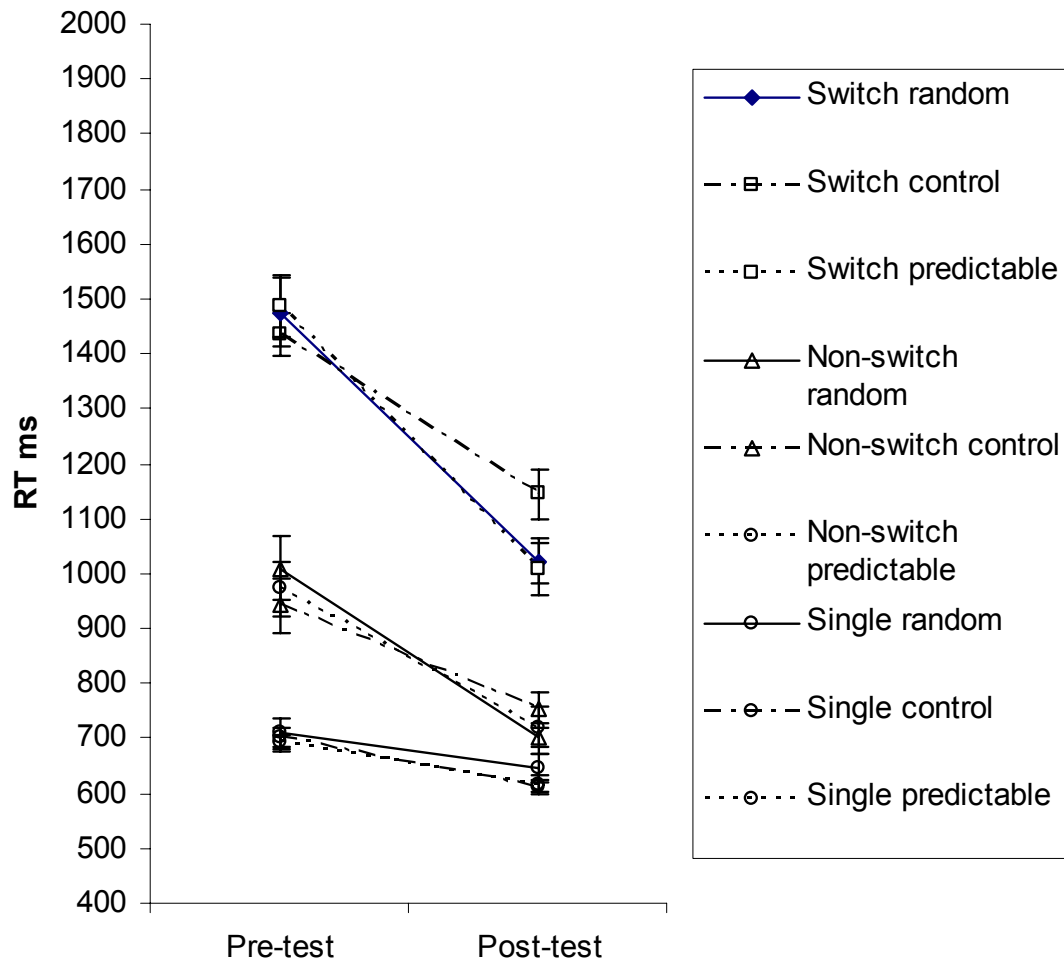


Figure 3b

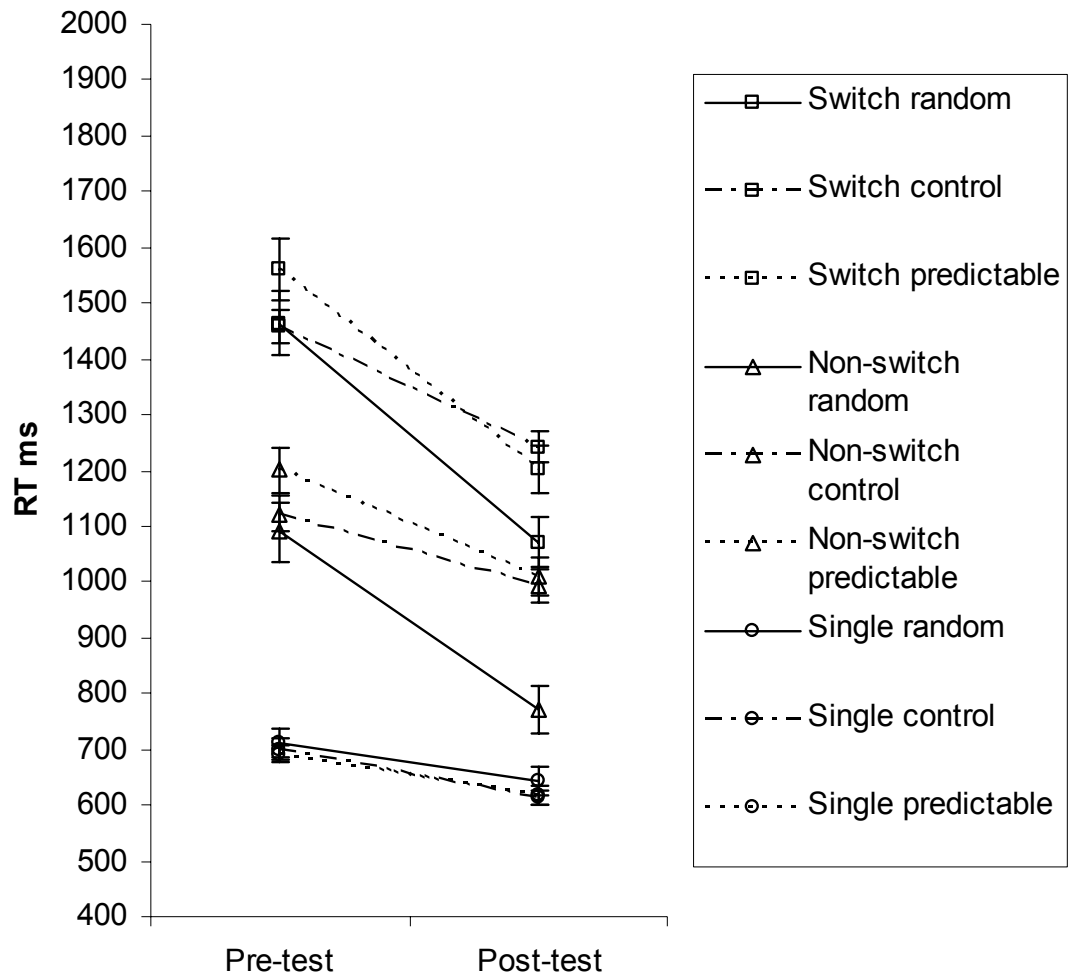


Figure 4a

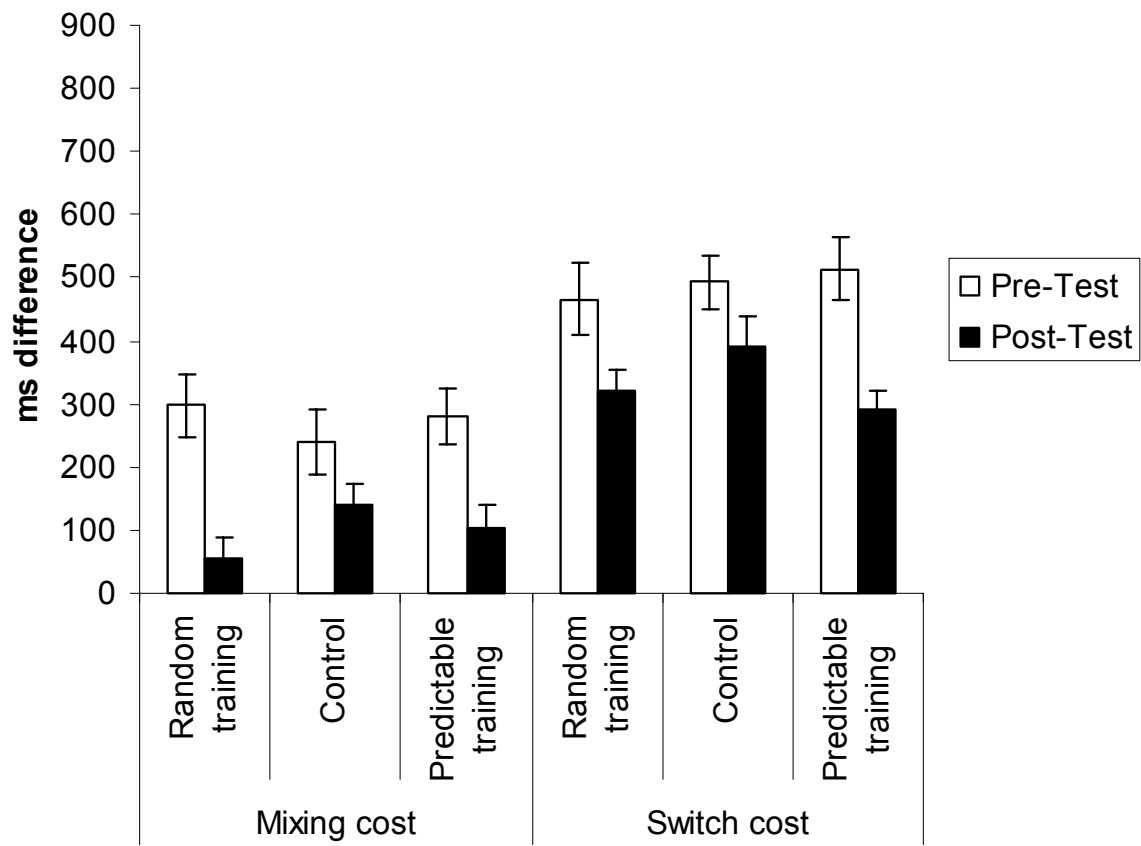


Figure 4b

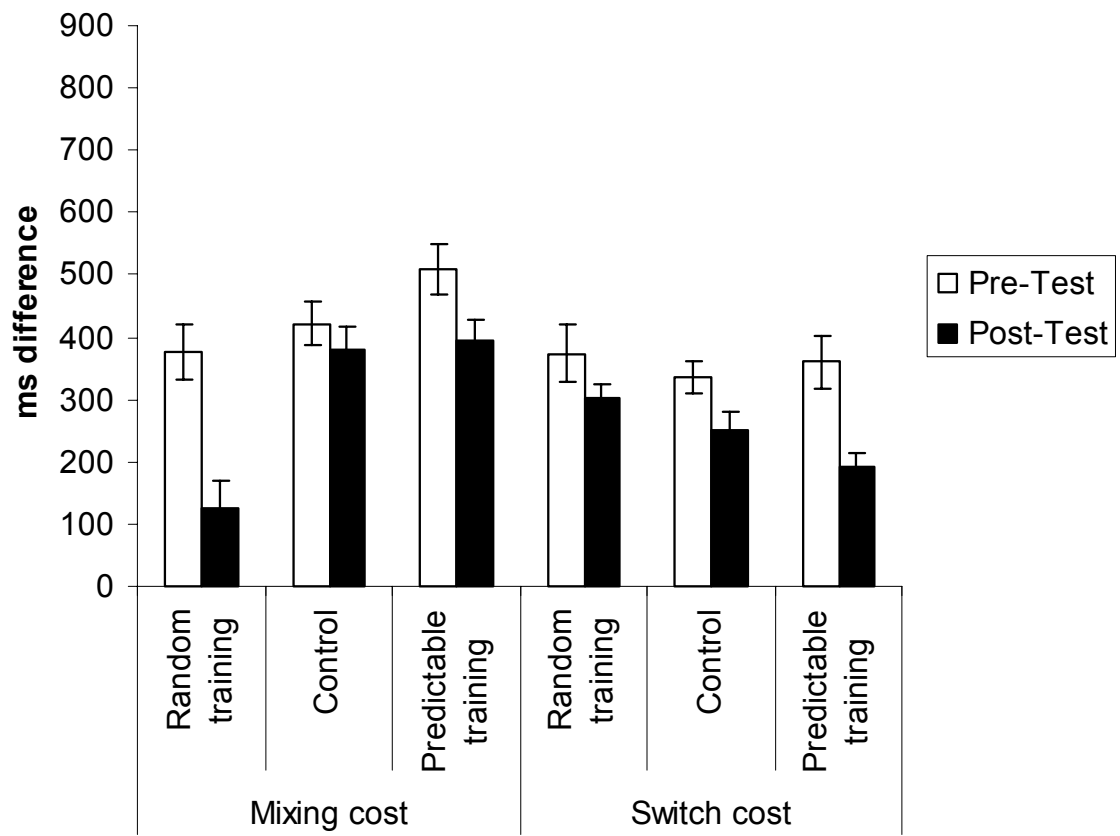


Figure 5a

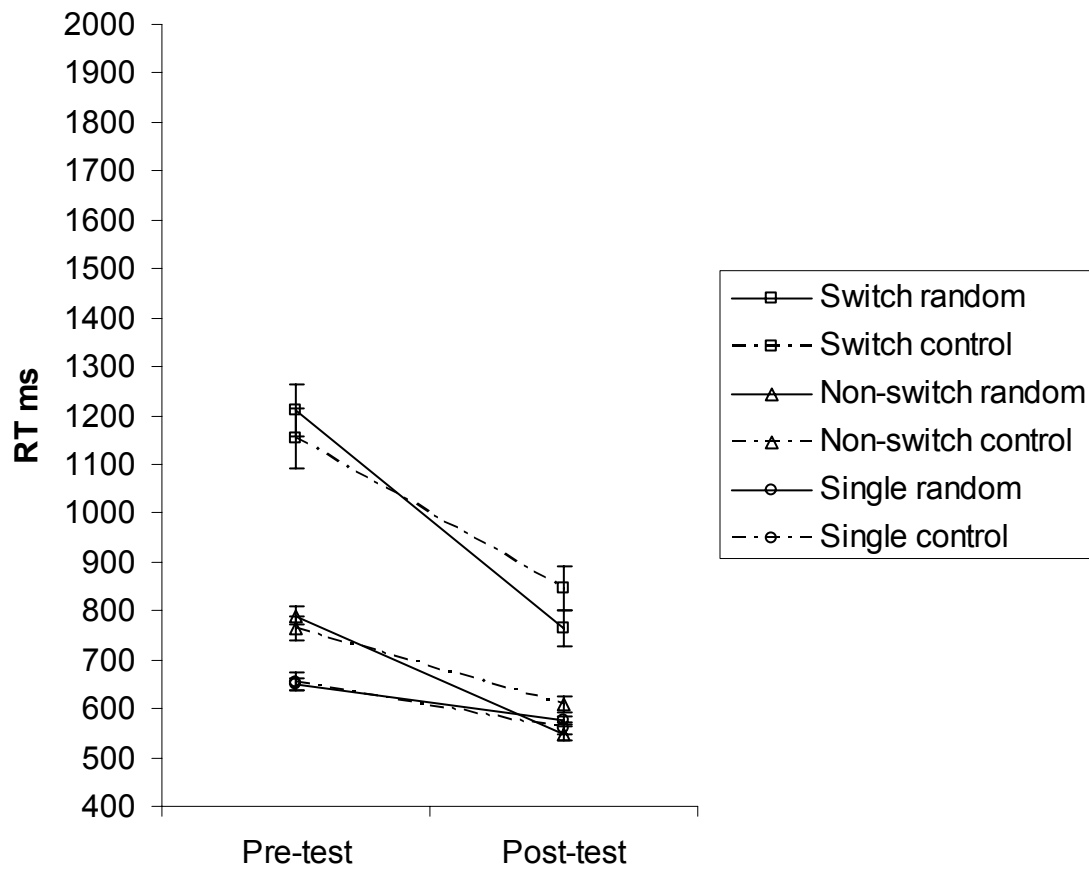


Figure 5b

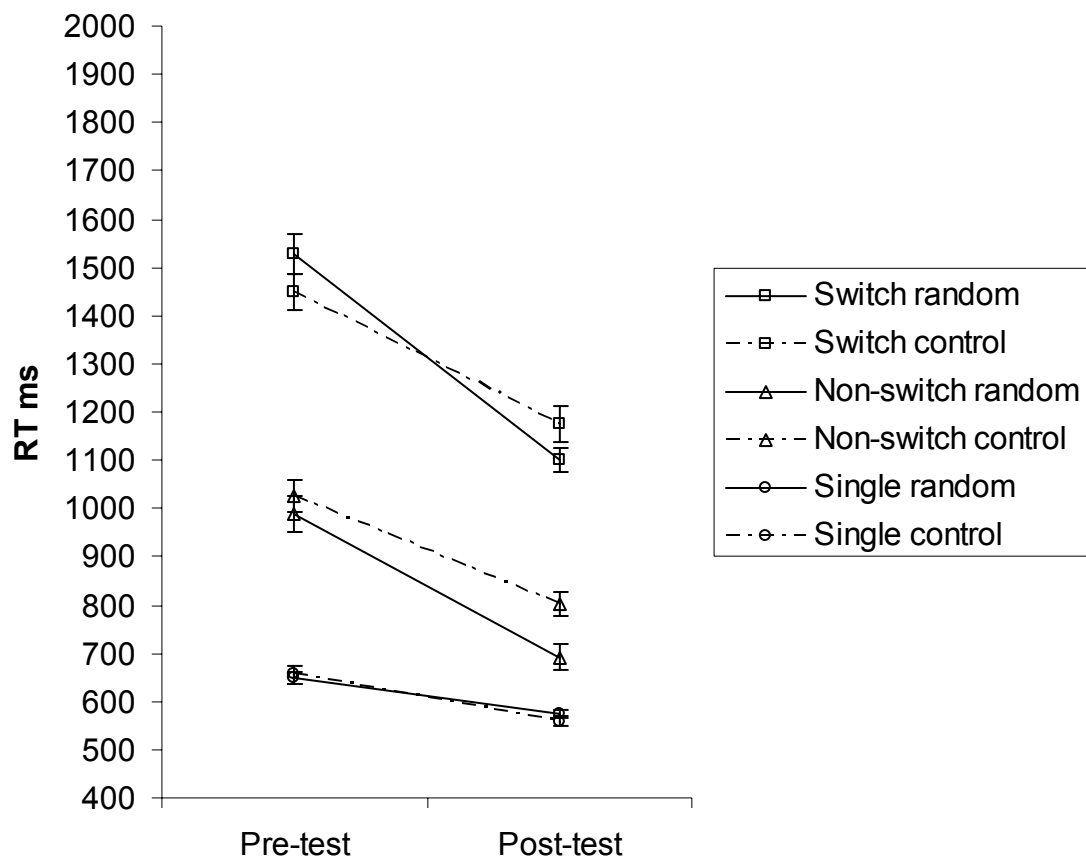


Figure 6a

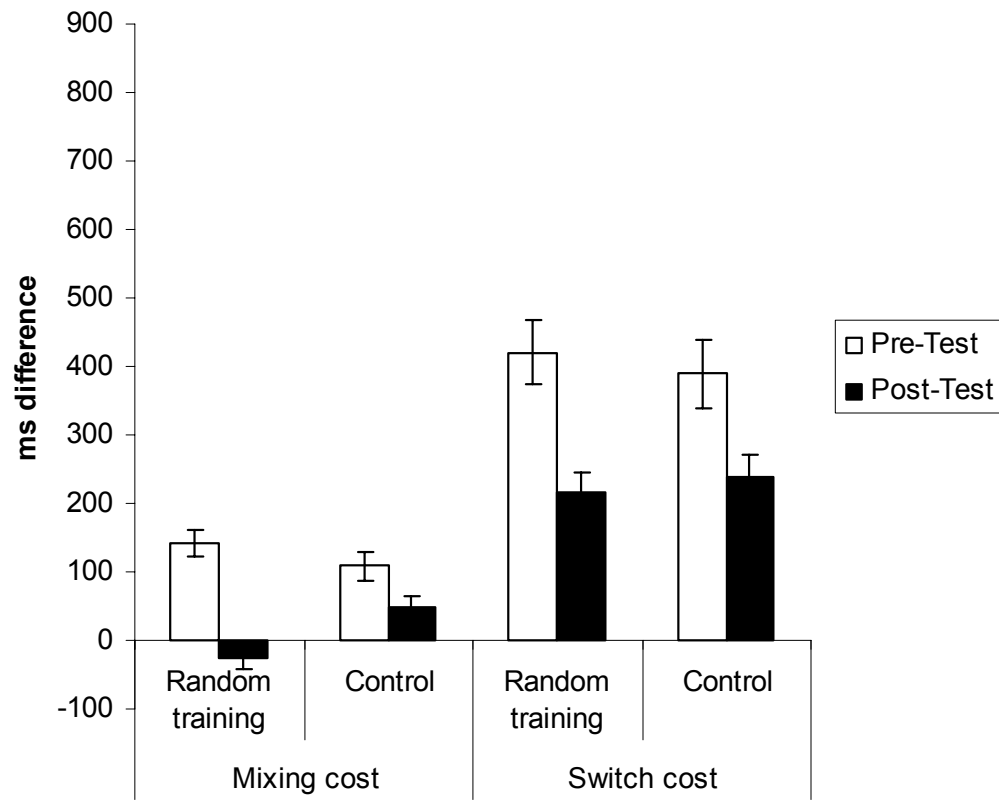


Figure 6b

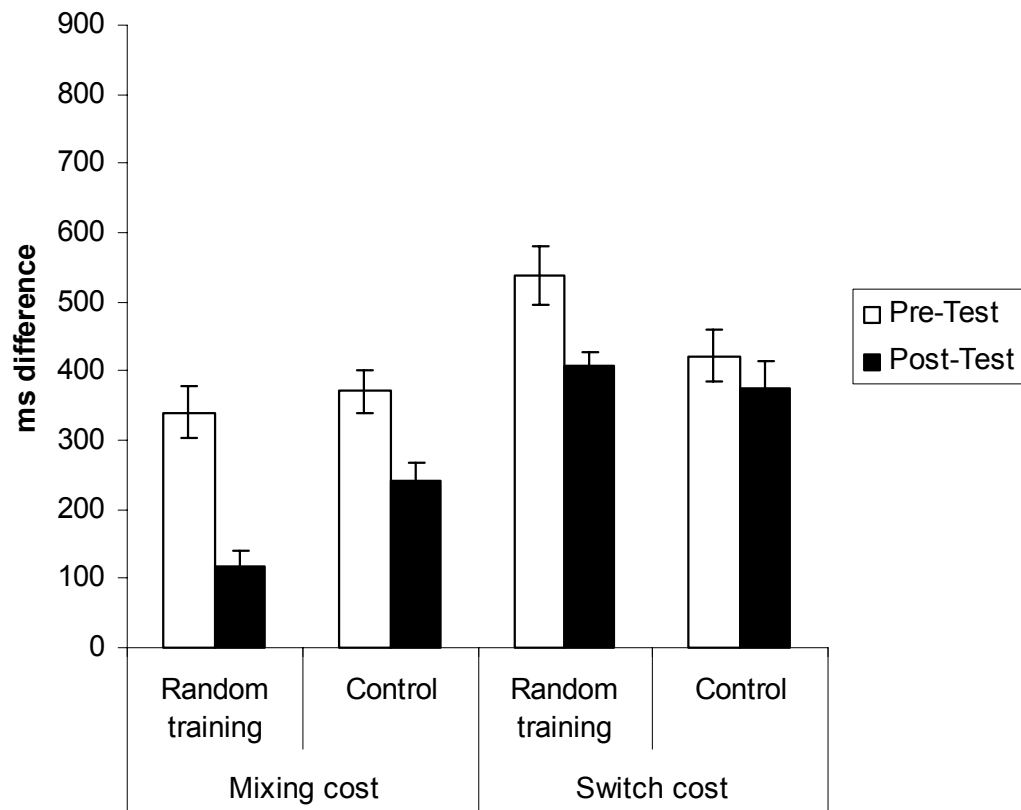


Figure 7a

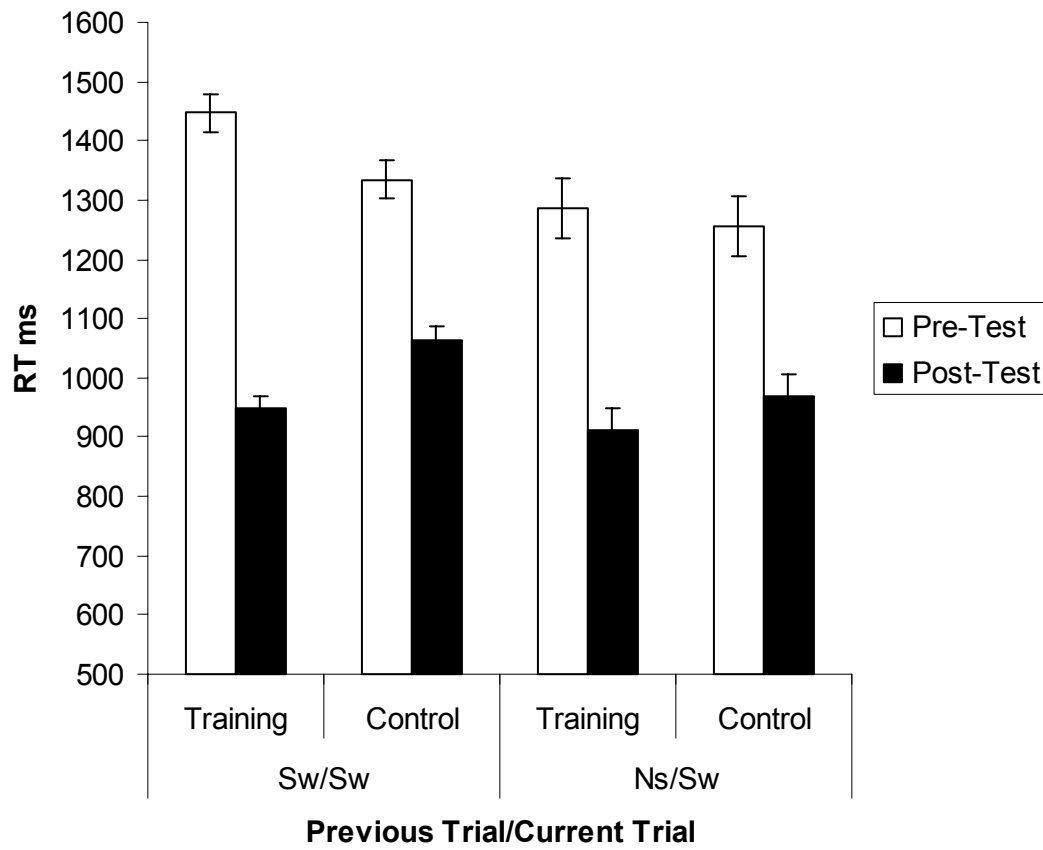


Figure 7b

